

5-2 Diagonalization

$$A = P^{-1}AP$$

$$B = P^{-1}AP$$

Key Points:

If A is $n \times n$ matrix, the following statements are equivalent.

- a) A is Diagonalizable
- b) A has Linearly independent eigenvectors.

Matrix is Diagonalizable iff there are three eigenvectors.

In a matrix of size 3×3 if there are three eigenvalue then matrix is always diagonalizable but in case of 2 eigenvalue it can be diagonalizable or can't be (depending on how many eigenvectors it has)

$$P^{-1}AP = \text{Diagonal Matrix}$$

$$A = PDP^{-1}$$

$$A = (PDP^{-1})^{100}$$

$$A = PD^{100}P^{-1}$$

Example: 1 (Find Matrix P that Diagonalizes a Matrix A)

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

1. Eigenvalues:-

$$\det(\lambda I - A) = 0$$

$$\lambda I - A =$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda - 2 & -1 \\ -1 & 0 & \lambda - 3 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda - 2 & -1 \\ -1 & 0 & \lambda - 3 \end{bmatrix} = 0$$

Expand along C_2

$$0 + (-1)^{2+2}(\lambda - 2) \begin{vmatrix} \lambda & 2 \\ -1 & \lambda - 3 \end{vmatrix} + 0 = 0$$

$$(\lambda - 2) [\lambda(\lambda - 3) + 2] = 0$$

$$(\lambda - 2)(\lambda^2 - 3\lambda + 2) = 0 \rightarrow \text{characteristic Equation}$$

$$\lambda - 2 = 0$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$\lambda^2 - 2\lambda - \lambda + 2 = 0$$

$$\lambda(\lambda - 2) - 1(\lambda - 2) = 0$$

$$(\lambda - 1)(\lambda - 2) = 0$$

$$\lambda - 1 = 0, \lambda - 2 = 0$$

$$\lambda = 1, \lambda = 2$$

2. Eigen Vectors:-

$$(\lambda I - A) X = 0$$

$$\begin{bmatrix} \lambda - 0 & 2 \\ -1 & \lambda - 2 & -1 \\ -1 & 0 & \lambda - 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 1$

$$= \begin{bmatrix} 1 & 0 & 2 & | & 0 \\ -1 & -1 & -1 & | & 0 \\ -1 & 0 & -2 & | & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 2 & | & 0 \\ 0 & -1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \begin{array}{l} R_2 + R_1 \\ R_3 + R_1 \end{array}$$

$$x_1 + 2x_3 = 0$$

$$x_1 = -2t$$

$$-x_2 + x_3 = 0$$

$$x_2 = t$$

Leading var = x_1, x_2 let $x_3 = t$

Free var = x_3

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2t \\ t \\ t \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} t$$

$$E_1 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

For $\lambda = 2$

$$= \begin{bmatrix} 2 & 0 & 2 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \end{bmatrix} \begin{matrix} R_1 \\ 2 \\ 2 \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_2 + R_1 \\ R_3 + R_1 \\ \end{matrix}$$

Leading var = x_1 Let

Free var = x_2, x_3 $x_2 = t, x_3 = s$

$$x_1 + x_3 = 0$$

$$x_1 = -s$$

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$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ t \\ s \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} + \begin{bmatrix} -5 \\ 0 \\ s \end{bmatrix}$$

$$X = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$E_2 = \left\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\rangle$$

3. Diagonal Matrix P

$$P = \begin{bmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

verify

$$D = P^{-1}AP$$

$$P^{-1} =$$

$$= \left[\begin{array}{ccc|ccc} -2 & 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ -2 & 0 & -1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

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$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & -1 & -1 & 2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array} \right] \begin{array}{l} R_1 + 2R_2 \\ R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \\ 0 & 2 & -1 & 1 & 2 & 0 \end{array} \right] R_2 \leftrightarrow R_3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 2 & -1 & 1 & 2 & 0 \end{array} \right] -R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{array} \right] \begin{array}{l} R_1 - R_2 \\ R_3 - 2R_2 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{array} \right] \begin{array}{l} R_2 + R_3 \\ R_1 - R_3 \end{array}$$

$$P^{-1} = \left[\begin{array}{ccc} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{array} \right]$$

$$P^{-1}A:-$$

$$= \left[\begin{array}{ccc|ccc} -1 & 0 & -1 & 0 & 0 & -2 \\ 1 & 1 & 1 & 1 & 2 & 1 \\ 1 & 0 & 2 & -1 & 0 & 3 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 0+0-1 & 0+0+0 & 2+0-3 \\ 0+1+1 & 0+2+1 & -2+1+3 \\ 0+0+2 & 0+0+0 & -2+0+6 \end{array} \right]$$

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$$= \begin{bmatrix} -1 & 0 & -1 \\ 2 & 2 & 2 \\ 2 & 0 & 4 \end{bmatrix}$$

$$D = P^{-1}AP$$

$$D = \begin{bmatrix} -1 & 0 & -1 \\ 2 & 2 & 2 \\ 2 & 0 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 2+0-1 & 0+0+0 & 1+0-1 \\ -4+2+2 & 0+2+0 & -2+0+2 \\ -4+0+4 & 0+0+0 & -2+0+4 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

↳ Diagonal Matrix
So verified

Example: 2 (Matrix that is not diagonalizable)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{bmatrix}$$

↳ First find eigenvalue

↳ Then find eigenvectors

↳ When we vectors there will be only
Find

two matrix eigenvectors and we discuss matrix is only diagonalizable when it has more than 3 eigenvectors.

Example : 3 (Recognizing Diagonalizability)

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

1. Eigenvalues

$$\det(\lambda I - A) = 0$$

$$\lambda I - A :-$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -4 & 17 & \lambda - 8 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -4 & 17 & \lambda - 8 \end{bmatrix} = 0$$

expand along R_1

λ	$\lambda - 1$	$-(-1)$	$0 - 1$	$+0 = 0$
	$17 \lambda - 8$		$-4 \lambda - 8$	

$$\lambda [\lambda(\lambda - 8) + 17] + 1(0 + 4) = 0$$

$$\lambda(\lambda^2 - 8\lambda + 17) + 4 = 0$$

$$\lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0 \rightarrow \text{char}$$

we have to find factors of 4 that satisfy characteristic eq.

$$\lambda = 4$$

$$(4)^3 - 8(4)^2 + 17(4) - 4 = 0$$

$$64 - 128 + 68 - 4 = 0$$

$$0 = 0$$

so $(\lambda - 4)$ is factor.

Use Synthetic Division

4	1	-8	17	-4
	↓	4	-16	4
	1	-4	1	0

$$\lambda = 4$$

$$\lambda^2 - 4\lambda + 1 = 0$$

$$a = 1, b = -4, c = 1$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{4 \pm \sqrt{16 - 4}}{2}$$

$$\lambda = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm \sqrt{2^2 \cdot 3}}{2}$$

$$\lambda = \frac{4 \pm 2\sqrt{3}}{2} = 2(2 \pm \sqrt{3})$$

$$\lambda = 2 + \sqrt{3}, \lambda = 2 - \sqrt{3}$$

2: Eigenvectors:

$$(\lambda I - A)X = 0$$

$$\begin{bmatrix} \lambda - 1 & 0 & 0 \\ 0 & \lambda - 1 & 0 \\ -4 & 17 & \lambda - 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 4$

$$= \begin{bmatrix} 4 & -1 & 0 & 0 \\ 0 & 4 & -1 & 0 \\ -4 & 17 & -4 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\frac{1}{4} & 0 & 0 \\ 0 & 4 & -1 & 0 \\ -4 & 17 & -4 & 0 \end{bmatrix} \begin{matrix} R_1 \times 4 \\ \\ \end{matrix}$$

$$= \begin{bmatrix} 1 & -\frac{1}{4} & 0 & 0 \\ 0 & 4 & -1 & 0 \\ 0 & 16 & -4 & 0 \end{bmatrix} \begin{matrix} \\ R_3 + 4R_2 \\ \end{matrix}$$

$$= \left[\begin{array}{ccc|c} 1 & -\frac{1}{4} & 0 & 0 \\ 0 & 1 & -\frac{1}{4} & 0 \\ 0 & 16 & -4 & 0 \end{array} \right] \quad R_2 \times \frac{1}{4}$$

$$= \left[\begin{array}{ccc|c} 1 & -\frac{1}{4} & 0 & 0 \\ 0 & 1 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad R_3 - 16R_2$$

Leading var = x_1, x_2 Let

Free var = x_3 $x_3 = t$

$$\begin{array}{l|l} x_1 - \frac{1}{4}x_2 = 0 & x_2 - \frac{1}{4}x_3 = 0 \\ x_1 - \frac{1}{4}(\frac{1}{4}t) = 0 & x_2 - \frac{1}{4}t = 0 \\ x_1 = \frac{1}{16}t = 0 & x_2 = \frac{1}{4}t \\ x_1 = \frac{1}{16}t & \end{array}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{16}t \\ \frac{1}{4}t \\ 1 \end{bmatrix} = t \begin{bmatrix} \frac{1}{16} \\ \frac{1}{4} \\ 1 \end{bmatrix}$$

we set $t=16$ to make further calculations simple.

$$X = \begin{bmatrix} 1 \\ 4 \\ 16 \end{bmatrix}$$

$$E_1 = \begin{bmatrix} 1 \\ 4 \\ 16 \end{bmatrix}$$

10/10/15

DATE: _____

For $\lambda = 2 + \sqrt{3}$

$$= \left[\begin{array}{ccc|c} 2+\sqrt{3} & -1 & 0 & 0 \\ 0 & 2+\sqrt{3} & -1 & 0 \\ -4 & 17 & 2+\sqrt{3}-8 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 2+\sqrt{3} & -1 & 0 & 0 \\ 0 & 2+\sqrt{3} & -1 & 0 \\ -4 & 17 & \sqrt{3}-6 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & -\frac{1}{2+\sqrt{3}} & 0 & 0 \\ 0 & 2+\sqrt{3} & -1 & 0 \\ -4 & 17 & -6+\sqrt{3} & 0 \end{array} \right] \quad \begin{array}{l} R_1 \\ \hline 2+\sqrt{3} \end{array}$$

Simplify Row 1.

$$\frac{-1}{2+\sqrt{3}} = \frac{-1}{2+\sqrt{3}} \times (2-\sqrt{3}) = \frac{-2+\sqrt{3}}{4-3} = -2+\sqrt{3}$$

$$= \left[\begin{array}{ccc|c} 1 & -2+\sqrt{3} & 0 & 0 \\ 0 & 2+\sqrt{3} & -1 & 0 \\ -4 & 17 & -6+\sqrt{3} & 0 \end{array} \right] \quad \begin{array}{l} 17-8+4\sqrt{3} \\ 9+4\sqrt{3} \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -2+\sqrt{3} & 0 & 0 \\ 0 & 2+\sqrt{3} & -1 & 0 \\ 0 & 9+4\sqrt{3} & -6+\sqrt{3} & 0 \end{array} \right] \quad \begin{array}{l} R_3 + 4R_1 \\ \hline \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -2+\sqrt{3} & 0 & 0 \\ 0 & 1 & -\frac{1}{2+\sqrt{3}} & 0 \\ 0 & 9+4\sqrt{3} & -6+\sqrt{3} & 0 \end{array} \right] \quad \begin{array}{l} R_2 \\ \hline 2+\sqrt{3} \end{array}$$

Simplify Row 2

$$= \frac{-1}{2+\sqrt{3}} = \frac{-1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{-2+\sqrt{3}}{4-3} = -2+\sqrt{3}$$

$$\begin{bmatrix} 1 & -2+\sqrt{3} & 0 & 0 \\ 0 & 1 & -2+\sqrt{3} & 0 \\ 0 & 9+4\sqrt{3} & -6+\sqrt{3} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2+\sqrt{3} & 0 & 0 \\ 0 & 1 & -2+\sqrt{3} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 - (9+4\sqrt{3})R_2$$

$$\begin{aligned} & -6+\sqrt{3} - (9+4\sqrt{3})(-2+\sqrt{3}) \\ & = -6+\sqrt{3} + 18 - 9\sqrt{3} + 8\sqrt{3} - 4\sqrt{3}^2 \\ & = -6+\sqrt{3} + 18 - 9\sqrt{3} + 8\sqrt{3} - 12 \\ & = -18 + 18 + \sqrt{3} - \sqrt{3} \\ & = 0 \end{aligned}$$

$$\begin{aligned} & -6+\sqrt{3} - (9+4\sqrt{3})(-2+\sqrt{3}) \\ & -6+\sqrt{3} \end{aligned}$$

Leading var: x_1, x_2 Let .

Free var = x_3 $x_3 = t$

$$\begin{aligned} x_1 + (-2+\sqrt{3})x_2 &= 0 & x_2 + (-2+\sqrt{3})x_3 &= 0 \\ x_1 + (-2+\sqrt{3})(2-\sqrt{3})t &= 0 & x_2 &= (2-\sqrt{3})t \\ x_1 + (-4+2\sqrt{3}+2\sqrt{3}-3)t &= 0 \\ x_1 &= (7-4\sqrt{3})t \end{aligned}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} (7-4\sqrt{3})t \\ (2-\sqrt{3})t \\ t \end{bmatrix} = t \begin{bmatrix} 7-4\sqrt{3} \\ 2-\sqrt{3} \\ 1 \end{bmatrix}$$

$$E_2 = \left\langle \begin{bmatrix} 7-4\sqrt{3} \\ 2-\sqrt{3} \\ 1 \end{bmatrix} \right\rangle$$

if we take $t = 7 + 4\sqrt{3}$

$$E_2 = \begin{pmatrix} 1 \\ 2 + \sqrt{3} \\ 7 + 4\sqrt{3} \end{pmatrix}$$

QUESTION 10

DATE: _____

For $\lambda = 2 - \sqrt{3}$

$$\begin{bmatrix} 2 - \sqrt{3} & -1 & 0 & 0 \\ 0 & 2 - \sqrt{3} & -1 & 0 \\ -4 & 17 & -6\sqrt{3} & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{2} - \sqrt{3} & 0 & 0 \\ 0 & 2 - \sqrt{3} & -1 & 0 \\ -4 & 17 & -6\sqrt{3} & 0 \end{bmatrix} \quad \begin{array}{l} R_1 \\ 2 - \sqrt{3} \end{array}$$

Simplify Row. 1

$$\begin{array}{l} -1 = -1 \times (2 + \sqrt{3}) = -2 - \sqrt{3} = -2 - \sqrt{3} \\ 2 - \sqrt{3} \quad 2 - \sqrt{3} \quad (2 + \sqrt{3}) \quad 4 - 3 \end{array}$$

$$\begin{bmatrix} 1 & -2 - \sqrt{3} & 0 & 0 \\ 0 & 2 - \sqrt{3} & -1 & 0 \\ -4 & 17 & -6\sqrt{3} & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 - \sqrt{3} & 0 & 0 \\ 0 & 2 - \sqrt{3} & -1 & 0 \\ 0 & 9 - 4\sqrt{3} & -6\sqrt{3} & 0 \end{bmatrix} \quad \begin{array}{l} R_1 \\ R_1 - 4R_2 \end{array}$$

$$\begin{bmatrix} 1 & -2 - \sqrt{3} & 0 & 0 \\ 0 & 1 & -2 - \sqrt{3} & 0 \\ 0 & 9 - 4\sqrt{3} & -6\sqrt{3} & 0 \end{bmatrix} \quad \begin{array}{l} R_2 \\ 2 - \sqrt{3} \end{array}$$

~~Ex 1~~

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$$= \left[\begin{array}{ccc|c} 1 & -2-\sqrt{3} & 0 & 0 \\ 0 & 1 & -2-\sqrt{3} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad R_3 - (9-4\sqrt{3})R_2$$

Leading var = x_1, x_2 Let

Free var = x_3 $x_3 = t$

$$\begin{aligned} x_1 + (-2-\sqrt{3})x_2 &= 0 & x_2 + (-2-\sqrt{3})x_3 &= 0 \\ x_1 + (-2-\sqrt{3})(2+\sqrt{3})t &= 0 & x_2 &= (2+\sqrt{3})t \\ x_1 + (-4-2\sqrt{3}-2\sqrt{3}-3)t &= 0 \\ x_1 + (-7-4\sqrt{3})t &= 0 \\ x_1 &= (7+4\sqrt{3})t \end{aligned}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} (7+4\sqrt{3})t \\ (2+\sqrt{3})t \\ t \end{bmatrix} = t \begin{bmatrix} 7+4\sqrt{3} \\ 2+\sqrt{3} \\ 1 \end{bmatrix}$$

$$E_3 = \left\langle \begin{bmatrix} 7+4\sqrt{3} \\ 2+\sqrt{3} \\ 1 \end{bmatrix} \right\rangle$$

if we take $t = 7-4\sqrt{3}$

$$E_3 = \left\langle \begin{bmatrix} 1 \\ 2+\sqrt{3} \\ 7-4\sqrt{3} \end{bmatrix} \right\rangle$$

so P

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 2+\sqrt{3} & 2-\sqrt{3} \\ 16 & 7+4\sqrt{3} & 7-4\sqrt{3} \end{bmatrix}$$

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2+\sqrt{3} & 0 \\ 0 & 0 & 2-\sqrt{3} \end{bmatrix}$$

↳

Can be Found Using → Proper Way

$$D = P^{-1}AP$$

Short Cut

put eigenvalues in diagonal matrix and that resultant will be find answers

$$D = P^{-1}AP$$

P^{-1} :-

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 4 & 2+\sqrt{3} & 2-\sqrt{3} & | & 0 & 1 & 0 \\ 16 & 7+4\sqrt{3} & 7-4\sqrt{3} & | & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & -2+\sqrt{3} & -2-\sqrt{3} & | & -4 & 1 & 0 \\ 0 & -9+4\sqrt{3} & -9-4\sqrt{3} & | & -16 & 0 & 1 \end{bmatrix} \begin{array}{l} R_2 - 4R_1 \\ R_3 - 16R_1 \end{array}$$

Example : 4 (Diagonalizability of Triangular Matrix)

$$A = \begin{bmatrix} -1 & 2 & 4 & 0 \\ 0 & 3 & 1 & 7 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

The eigen values of triangular matrix are the entries on its main diagonal, Thus a triangular matrix with distinct entries on main diagonal is diagonalizable.

So A is diagonalizable with eigenvalues

$$\lambda_1 = -1, \lambda_2 = 3, \lambda_3 = 5, \lambda_4 = -2$$

Example : 5 (Eigenvalues and eigen vectors of matrix Powers)

Find A^7

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{bmatrix}$$

we will solve such type of q's in Exercise

Example 6: (Powers of Matrix)

Find A^3

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

1. Eigenvalues:

$$\det(\lambda I - A) = 0$$

$$\lambda I - A = 0$$

$$= \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda - 2 & -1 \\ -1 & 0 & \lambda - 3 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda - 2 & -1 \\ -1 & 0 & \lambda - 3 \end{bmatrix} = 0$$

Expand along C_2

$$0 + (-1)^{2+2}(\lambda - 2) \begin{vmatrix} \lambda & 2 \\ -1 & \lambda - 3 \end{vmatrix} + 0 = 0$$

$$(\lambda - 2) [\lambda(\lambda - 3) + 2] = 0$$

$$(\lambda - 2)(\lambda^2 - 3\lambda + 2) = 0$$

$$\lambda - 2 = 0$$

$$\lambda = 2$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$\lambda^2 - 2\lambda - \lambda + 2 = 0$$

$$\lambda(\lambda - 2) - 1(\lambda - 2) = 0$$

$$(\lambda - 1)(\lambda - 2) = 0$$

$$\lambda = 1, \lambda = 2$$

2. Eigen vectors

$$(\lambda I - A) X = 0$$

$$\begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda - 2 & -1 \\ -1 & 0 & \lambda - 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 1$

$$= \begin{bmatrix} 1 & 0 & 2 & 0 \\ -1 & -1 & -1 & 0 \\ -1 & 0 & -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_2 + R_1 \\ R_3 + R_1 \\ \end{matrix}$$

Leading var = x_1, x_2 Let

Free var = x_3

$$x_3 = t$$

$$x_1 + 2x_3 = 0$$

$$x_1 + 2t = 0$$

$$x_1 = -2t$$

$$-x_2 + x_3 = 0$$

$$x_2 = t$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$E_1 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

2015/11/10

For $\lambda = 2$

$$= \begin{bmatrix} 2 & 0 & 0 & | & 0 \\ -1 & 0 & -1 & | & 0 \\ -1 & 0 & -1 & | & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & | & 0 \\ -1 & 0 & -1 & | & 0 \\ -1 & 0 & -1 & | & 0 \end{bmatrix} \quad R_1 \times \frac{1}{2}$$

$$= \begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \quad \begin{array}{l} R_2 + R_1 \\ R_3 + R_1 \end{array}$$

Leading Var = x_1 Let

Free Var $\rightarrow x_2, x_3$ $x_2 = t, x_3 = s$

$$x_1 + x_3 = 0$$

$$x_1 + s = 0$$

$$x_1 = -s$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -s \\ t \\ s \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} + \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix}$$

$$X = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$E_2 = \left\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\rangle$$

As three eigenvector so matrix

is diagonalizable. $D = P^{-1}AP$ short cut (Put Eigen value in Diagonal of Diagonal Matrix)

So P

$$P = \begin{bmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$D = P^{-1}AP \Rightarrow A = PDP^{-1} \Rightarrow A = PDP^{-1}$$

$$P^{-1}$$

$$P^{-1} = \begin{bmatrix} -2 & 0 & -1 & | & 1 & 0 & 0 \\ 1 & 1 & 0 & | & 0 & 1 & 0 \\ 1 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 & | & 0 & 1 & 0 \\ -2 & 0 & -1 & | & 1 & 0 & 0 \\ 1 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

ADDITION

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$$= \left[\begin{array}{ccc|ccc} -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & -1 & 1 & 2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array} \right] \begin{array}{l} R_2 + 2R_1 \\ R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \\ 0 & 2 & -1 & 1 & 2 & 0 \end{array} \right] R_2 \leftrightarrow R_3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 2 & -1 & 1 & 2 & 0 \end{array} \right] -R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{array} \right] \begin{array}{l} R_1 - R_2 \\ R_3 - 2R_2 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{array} \right] \begin{array}{l} R_2 + R_3 \\ R_1 - R_3 \end{array}$$

$$P^{-1} = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{array} \right]$$

PD:

$$= \left[\begin{array}{ccc|ccc} -2 & 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 8192 & 0 \\ 1 & 0 & 1 & 0 & 0 & 8192 \end{array} \right]$$

Q. 1. (10.5)

DATE: _____

$$= \begin{bmatrix} -2+0+0 & 0+0-0 & 0+0-8192 \\ 1+0+0 & 0+8192+0 & 0+0+0 \\ 1+0+0 & 0+0+0 & 0+0+8192 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & -8192 \\ 1 & 8192 & 0 \\ 1 & 0 & 8192 \end{bmatrix}$$

$$A^3 = (PD^3)P^{-1}$$

$$A^3 = \begin{bmatrix} -2 & 0 & -8192 \\ 1 & 8192 & 0 \\ 1 & 0 & 8192 \end{bmatrix} \begin{bmatrix} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2+0-8192 & 0+0+0 & 2+0-16384 \\ -1+8192+0 & 0+8192+0 & -1+8192+0 \\ -1+0+8192 & 0+0+0 & -1+0+16384 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -8190 & 0 & -16382 \\ 8191 & 8192 & 8191 \\ 8191 & 0 & 16383 \end{bmatrix} \quad \text{Ans}$$

Exercise : 5.2

Question : 17-18

Compute A^{10} See Example 6
For solution.Question : 19 (compute A'')see Example no 1 & Example
no. 6Question : 20 (a, b, c)

$$A = \begin{bmatrix} 1 & -2 & 8 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, P = \begin{bmatrix} 1 & -4 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

See Example 1 & Example

Note:- For negative powers
shift power to denominator D^{-100} Like Example
→ Assume.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$D^{-100} = \begin{bmatrix} \frac{1}{(2)^{100}} & 0 & 0 \\ 0 & \frac{1}{(2)^{100}} & 0 \\ 0 & 0 & \frac{1}{(2)^{100}} \end{bmatrix}$$

21-Question:

Find A^n if n is positive integer and

$$A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

Assume n whatever you want like (11, 13, 100) and the follow method of example.

Question: 9

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix}$$

λ

$$\det(\lambda I - A) = 0$$

$$\lambda I - A =$$

$$= \begin{bmatrix} \lambda - 4 & 0 & -1 \\ -2 & \lambda - 3 & -2 \\ -1 & 0 & \lambda - 4 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda-4 & 0 & -1 \\ -2 & \lambda-3 & -2 \\ -1 & 0 & \lambda-4 \end{bmatrix} = 0$$

expand along C_2

$$0 + (-1)^{2+2} (\lambda-3) \begin{vmatrix} \lambda-4 & -1 \\ -1 & \lambda-4 \end{vmatrix} + 0 = 0$$

$$(\lambda-3) [(\lambda-4)^2 + 1] = 0$$

$$(\lambda-3)(\lambda^2 - 8\lambda + 16 + 1) = 0$$

$$(\lambda-3) = 0$$

$$\lambda = 3$$

$$\lambda^2 - 8\lambda + 15 = 0$$

$$\lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\lambda(\lambda-5) - 3(\lambda-5) = 0$$

$$(\lambda-3)(\lambda-5) = 0$$

$$\lambda = 3, \lambda = 5$$

b-

For $\lambda = 3$

$$(\lambda I - A)$$

$$= \begin{bmatrix} -1 & 0 & -1 \\ -2 & 0 & -2 \\ -1 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ -2 & 0 & -2 \\ -1 & 0 & -1 \end{bmatrix} \xrightarrow{-R_1}$$

Q 1. (M, 1, 1, 1, 5)

DATE: _____

$$= \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 + R_1 \end{array}$$

As number of non zero row is 1
So Rank = 1 for $\lambda = 3$

For $\lambda = 5$

$$\lambda I - A = \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ -2 & 2 & -2 & 0 \\ -1 & 0 & 1 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 2 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_2 + 2R_1 \\ R_3 + R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

As number of non-zero row is 2
So Rank = 2 for $\lambda = 5$

C-

1. First Way:

Find eigenvectors and if they comes out to be 3 (greater than 2) then matrix will be Diagonalizable.

2. Second Way

A 3×3 matrix is Diagonalizable iff sum of dimension of its eigenspace equals 3.

$$\text{Dimension of Eigenspace} = n - \text{rank}(\lambda I - A)$$

For $\lambda = 3$.

$$\text{Dimen} = 3 - 1$$

$$\text{Dimen} = 2$$

For $\lambda = 5$

$$\text{Dimen} = 3 - 2$$

$$\text{Dimen} = 1$$

Sum:

$$\text{Total} = 2 + 1$$

$$\text{Total} = 3$$

So Matrix is Diagonalizable.

Question: 10

Follow Direction in Exercise 9
for matrix.

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

same as $q:10$.